Satyam Awasthi

Closed-Loop Brain-Computer Interfaces | Sensing, Decoding & Control

satyam.aw@yahoo.com satyam-aw.github.io satyam-aw satyam-awasthi

Research Profile

I develop methods for neural decoding and reliable closed-loop brain–computer interfaces. My current research spans real-time EEG interfaces, motor decoding, sensory feedback, and robust control under uncertainty.

Education

University of California, Santa Barbara Santa Barbara, CA
M.S. in Computer Science Sep 2021 – Mar 2023
GPA: 3.91/4.00. Four Eyes Laboratory; advisors: Tobias Höllerer and Michael Beyeler.

Indian Institute of Technology, Kharagpur Kharagpur, India
B.Tech. in Electrical Engineering Jul 2016 – Jul 2020
GPA: 8.98/10.0. Minor in Computer Science.

Current Research Projects

SSVEP-Based EEG Navigation

2026 – present

Guidance: Dr. Camille Gontier (Inria).

- Built a four-channel BioAmp/MG24 EEG system, a custom electrode cap, and photodiode timing tools.
- Tested CCA decoding with one and four targets. Used recorded EEG to tune window length, harmonics, and decision thresholds.
- Integrated live CCA decoding with a C++/OpenGL maze and tested the pipeline end to end.

EEG-to-Kinematics Decoding

2026 – present

Guidance: Prof. Slawomir Nasuto (University of Reading).

- Investigate how reliably EEG can reconstruct wrist movement during grasping and lifting, and whether separate decoders could improve predictions for specific movement stages.
- Compared six decoding models. Strong performance within one participant did not transfer reliably to 11 others without adaptation.
- Tested how EEG timing and input duration affect wrist decoding across 12 participants. Longer windows with more averaging reduced held-out prediction error for a convolutional model in all 12, by 17% on average.

Error-Related EEG During Mouse Movements

2026 – present

Guidance: Dr. Serafeim Perdikis (University of Essex); collaboration with Mushfika Sultana.

- Cluster-based permutation tests on reaches matched for target entry identified significant EEG differences after target displacement.
- Tested whether EEG could distinguish displaced from non-displaced trials on held-out runs within each participant and session, achieving 76.3% mean balanced accuracy across 23 non-pilot participants.

Sensory Feedback for Bidirectional Neural Interfaces

2026 – present

Guidance: Prof. Mohsen Rakhshan (University of Central Florida).

- Studying how peripheral stimulation waveforms relate to EEG responses and perceived sensation.
- Testing whether single EEG epochs distinguish five waveforms across participants while checking for stimulation and task confounds.

Robust Output-Feedback MPC

Jul 2026 – present

Guidance: Dr. Johannes Köhler (Imperial College London).

- Developing joint bounds for state-estimation and tracking errors under state-dependent uncertainty.
- Reduced the final observer radius by 6.68% in a synthetic bound-propagation experiment. Closed-loop validation is ongoing.

Earlier Research Experience

Four Eyes Laboratory, UC Santa Barbara Jan 2022 – Mar 2023

Graduate Researcher - EyeTTS

Advisor: Prof. Tobias Höllerer; co-advisor: Prof. Michael Beyeler.

- Built a Magic Leap 1 eye-tracking study and a Python pipeline for calibration and analysis during mixed-reality locomotion.
- Analysed gaze error and timing during target, head, and body movement. First author on two poster papers.

Undergraduate Robotics Research, IIT Kharagpur 2016 – 2018

Perception, sensor fusion, and navigation for the Eklavya autonomous ground vehicle.

Publications

Eye Tracking Performance in Mobile Mixed Reality

Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, Tobias Höllerer

IEEE VR Workshops (VRW) (Workshop poster), 2024

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, Tobias Höllerer

IEEE ISMAR Adjunct (Workshop poster), 2023

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, Bivas Mitra

COMSNETS, 2022

Skills

Programming & scientific computing: Python, C++, MATLAB, NumPy/SciPy, PyTorch

Neural signal processing: EEG preprocessing, spectral analysis, digital filtering, CCA, time-series decoding

Control & estimation: robust output-feedback MPC, contraction-based methods, uncertainty propagation

EEG & experimental systems: BioAmp EEG acquisition, photodiode timing, OpenGL

Teaching

Graduate Teaching Assistant, UC Santa Barbara (Sep 2021–Mar 2023). Taught six quarters across CMPSC 16, 24, 184, and 189A/B, mentored capstone teams, and supported 300+ students.

Industry Experience

Intuit — Software Engineer 2 · Mar 2024–Jan 2026

Yahoo — Software Dev Engineer II (IC3) · Apr–Nov 2023

Coupang — Software Engineering Intern · Jun–Sep 2022

Samsung Research Institute — Software Engineer (CL 2) · Sep 2020–Sep 2021

Academic Service

- Peer Reviewer, ACM UIST 2024 (interactive systems, spatial tracking, and interface hardware).